

Hypothesis 2 - Effect of Negative Emotion on Working-Memory Performance

Confirmatory Hypothesis Report

Hypothesis

Negative emotional induction leads to poorer working-memory performance relative to the neutral emotional condition.

Method

A confirmatory trial-level analysis was conducted to determine whether negative emotional induction impaired working-memory performance relative to the neutral condition. The analytical sample included 23 participants.

The primary outcomes were response accuracy and reaction time. Accuracy was analyzed using a generalized estimating equation (GEE) model with a binomial distribution, logit link, exchangeable within-participant correlation structure, and robust standard errors. Emotional condition was entered as the predictor, with the neutral condition specified as the reference category. The accuracy analysis included 11,880 trial-level observations.

Reaction time was analyzed among valid correct-response trials only. Because reaction-time values were positively skewed, reaction time was log-transformed before modeling. A linear mixed-effects model was fitted with emotional condition as a fixed effect and a participant-specific random intercept. Maximum-likelihood estimation was used. The reaction-time analysis included 9,533 observations.

Model coefficients from the binomial GEE were exponentiated and interpreted as odds ratios. Coefficients from the log reaction-time model were exponentiated and converted to percentage changes relative to the neutral condition. Inverse efficiency was additionally examined as a secondary indicator of the combined speed-accuracy pattern. Statistical significance was evaluated using a two-sided alpha level of .05, and 95% confidence intervals are reported.

Results

Accuracy

Mean trial-level accuracy was 83.8% in the negative condition and 85.0% in the neutral condition. The descriptive difference was -1.2 percentage points.

Relative to the neutral condition, negative emotion was associated with a small reduction in the estimated odds of a correct response ($\beta = -0.081$, $SE = 0.065$, $z = -1.25$, $p = .210$). The corresponding odds ratio was 0.922 (95% CI [0.81, 1.05]).

The confidence interval included the null value of 1.00, and the comparison was not statistically significant. Therefore, the analysis did not provide sufficient evidence that negative emotional induction reduced response accuracy relative to the neutral condition.

Table 1. Descriptive accuracy under neutral and negative emotion.

Condition	Participants	Trials	Correct	Accuracy	SD	Median
Negative	22	5940	4978	83.8%	0.368	1.0
Neutral	23	5940	5049	85.0%	0.357	1.0

Table 2. Binomial GEE estimate for response accuracy.

Contrast	Beta	OR	95% CI	p
Negative vs Neutral	-0.081	0.922	[0.81, 1.05]	= .210

Reaction Time

Mean reaction time was 740.2 ms in the negative condition and 966.6 ms in the neutral condition.

The negative emotional condition was significantly associated with lower log reaction time relative to the neutral condition (beta = -0.268, SE = 0.015, z = -17.37, p < .001).

After back-transformation, reaction time under negative emotion was estimated to be 23.5% lower than under the neutral condition (95% CI [-25.8%, -21.1%]). Thus, participants responded significantly faster, rather than more slowly, during the negative emotional condition.

Table 3. Descriptive reaction time under neutral and negative emotion.

Condition	Participants	Trials	Mean RT	SD	Median RT	Mean log RT
Negative	22	4704	740.25	744.23	433.50	6.25
Neutral	23	4829	966.58	896.36	630.00	6.51

Table 4. Mixed-effects model estimate for log reaction time.

Contrast	Beta	RT change	95% CI	p
Negative vs Neutral	-0.268	-23.5%	[-25.8%, -21.1%]	< .001

Processing Efficiency

Inverse efficiency was examined as a secondary measure combining reaction speed and response accuracy. Mean IES was 875.11 under negative emotion and 1170.40 under the neutral condition.

The paired comparison yielded a mean difference of -295.29, t = -4.74, p < .001. Negative emotion was associated with significantly better combined processing efficiency.

Table 5. Paired inverse-efficiency comparison.

Comparison	N	Neutral IES	Negative IES	Difference	t	p
Negative vs Neutral	22	1170.40	875.11	-295.29	-4.74	< .001

Decision Rules

- H2 is supported if negative emotion is associated with significantly lower accuracy, significantly slower reaction time, or both.
- H2 is partially supported if only one impairment outcome is statistically supported or if there is clear evidence of a speed-accuracy trade-off.
- H2 is not supported if negative emotion does not significantly reduce accuracy and responses are not significantly slower than under neutral emotion.

Interpretation

Hypothesis 2 was not supported.

The hypothesis predicted that negative emotional induction would impair working-memory performance relative to the neutral condition.

Although response accuracy was descriptively lower under negative emotion, the difference was small and statistically non-significant. Therefore, there was no reliable evidence that negative emotion decreased the probability of producing a correct response.

Reaction-time results were opposite to the predicted impairment pattern. Rather than responding more slowly, participants responded approximately 23.5% faster under negative emotion than under the neutral condition.

The combination of significantly faster responses and a small, non-significant reduction in accuracy may be descriptively compatible with a shift toward faster responding. However, it does not provide strong evidence of a speed-accuracy trade-off because the reduction in accuracy was not statistically supported.

Overall, negative emotional induction altered response speed but did not produce the expected deterioration in working-memory performance.

Hypothesis Assessment

Conclusion: Hypothesis 2 was not supported. Negative emotion did not significantly reduce response accuracy and was associated with faster, rather than slower, reaction times relative to the neutral condition.